9
Economics Associated with Implementation of Membrane Reactors

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9.1
Introduction

Recently, the need for *sustainable development* has been strongly recommended. Industrial production and innovation must satisfy the ever changing requirements of a modern society but are also expected to address preserving and improving the quality of life. Higher production capacities, increased safety, reduced equipment size, reduced energy and raw materials consumption, reduced waste production, reduced costs, improved yields and improved efficiencies, and remote control and automation are all goals to pursue for *sustainable growth* in the future.

In this scenario, membrane reactors can play an important role. They have intrinsic properties typical of membrane operations, such as modularity, no need for chemical additives, easy scale-up and control, high flexibility, and easy integration with conventional operations which make them good candidates for industrial production applications. Furthermore, membrane reactors are able to couple the reaction and the separation steps in a single device, reducing equipment size and avoiding the need for a separation step downstream. They also lead to higher yields and conversions and lower energy consumption compared to traditional reactors, representing a valid means for improving production efficiency. A study of costs associated with these systems and identification of the factors influencing their economics is, therefore, of crucial importance for defining their effective application at the industrial level.

The aim of this chapter is to give a critical analysis of parameters that must be taken into account each time an economic study on membrane reactors is performed. In particular, nonporous inorganic membrane reactors for reactions involving gases are considered.

Dense membrane reactors are discussed, emphasizing the most frequently investigated configurations and applications. Factors influencing economics are analyzed and, as a case study, results on the water-gas shift reaction in Pd-based membrane reactors are presented.

The water-gas shift reaction is used in refineries as a step to further reduce CO contained in the exit streams of methane reformers and to adjust the CO/H_2 ratio

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of the syngas produced. It is also useful in fuel cell applications, which require feed streams with very low CO content (usually, less than 10 ppm). Several projects are in progress worldwide on the use of membrane reactors for carrying out the water-gas shift reaction. It is, therefore, of interest to identify the main features as well as economic and engineering issues related to these systems. The last section of the chapter points out the type of research which needs to be pursued in order to apply membrane reactors in industrial applications.

9.2 Membrane Reactors

A membrane reactor can be viewed as a catalytic device containing a membrane. The membrane can be catalytically active or inert. In the former case, the reaction can take place within the membrane structure (if all the membrane is catalytic) or on the membrane surface (if only the membrane surface is catalytic). If the membrane is inert, the reaction usually occurs on catalytic particles packed close to the membrane (see Fig. 9.1(a) and (b)).

In membrane reactors, reaction and separation can be carried out simultaneously (Fig. 9.2), and it is possible to remove one of the products before it can further react and lead to undesired by-products. A reactant can be added in a con-

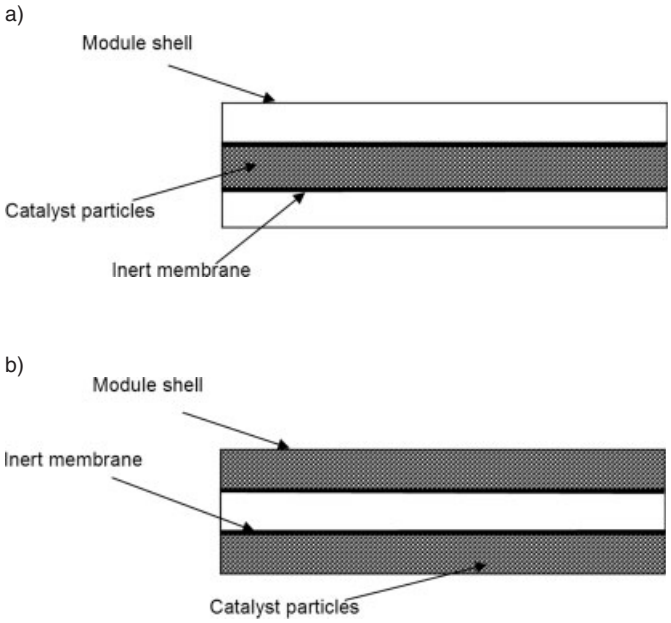


Figure 9.1 (a) Tubular inert membrane reactor with the catalyst packed in the lumen side. (b) Tubular inert membrane reactor with the catalyst packed in the shell side.

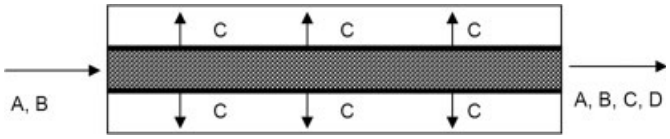


Figure 9.2 Reaction and separation carried out simultaneously in a membrane reactor (A, B, reactants; C, D, products).

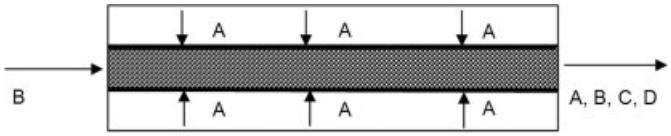


Figure 9.3 Controlled addition of a reactant through the membrane (A, B, reactants; C, D, products).

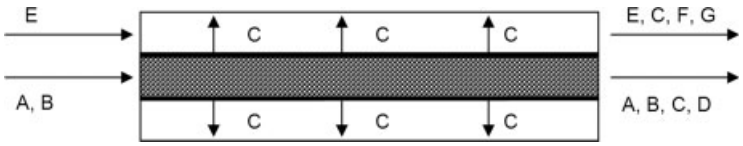


Figure 9.4 Coupling of two reactions in the same device (Reaction 1: A, B, reactants; C, D, products. Reaction 2: E, C, reactants; F, G, products).

trolled way (Fig. 9.3), and separate feeds of reactants can be employed. Two reactions can be carried out in the same device (Fig. 9.4). Catalyst loss can also be reduced when catalytic membranes are employed.

When membrane reactors are applied to separate a gaseous product from the reaction zone, the driving force for permeation can be obtained by reducing the partial pressure of the product at the permeate side using vacuum (Fig. 9.5), sending a sweep gas into the permeate (Fig. 9.6), or by increasing the partial pressure in the feed stream (Fig. 9.7).

The process efficiency depends on both the permeation and the reaction rates. Low permeation rates imply low removal, and, therefore, the membrane does not affect overall reaction performance significantly. If the permeation rate is higher than the reaction rate, the process is controlled by catalyst activity, and the mem-

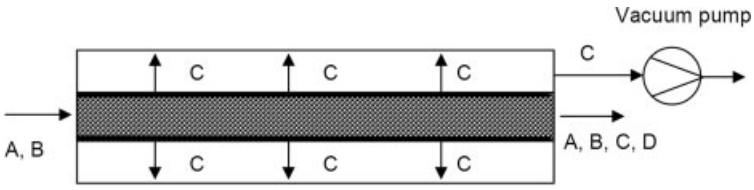


Figure 9.5 Driving force obtained by applying vacuum to the permeate side.

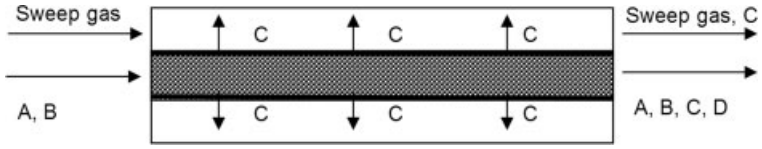


Figure 9.6 Driving force obtained by sending a sweep gas past the permeate side.

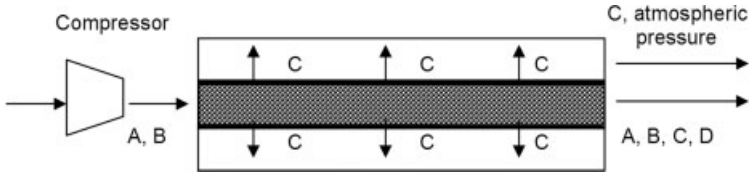


Figure 9.7 Driving force obtained by pressurizing the feed.

brane reactor is not well utilized. Ideally, the reaction rate should be as close as possible to the permeation rate, in order to remove the product as it is formed.

Generally, membrane materials with high selectivities and permeabilities are required in order to achieve good membrane reactor performance. Furthermore, for applications in chemical industry, membranes resistant to aggressive chemical environments and able to withstand high temperatures and pressures are required.

All these issues have increased the use of inorganic membranes. Dense membranes are usually preferred because of their higher selectivity with respect to porous ones. Typical dense membranes are made of perovskites or metals. In particular, many studies have been made on the use of Pd-based membrane reactors for carrying out hydrogen-based reactions [1–6]. When Pd or Pd alloys are employed, 100% selectivity to hydrogen is achieved, and pure hydrogen can be recovered at the permeate side (dehydrogenations) or can be fed in a controlled way through the membrane (hydrogenations). Often, silver or other metals belonging to Group VIIIB of the periodic table are added to the palladium lattice in order to significantly reduce hydrogen embrittlement. Furthermore, the addition of specific amounts of other metals in the Pd structure can lead to an increase of permeation rate. For example, by adding 20–30 wt% of silver, permeation rates up to 70% higher with respect to pure Pd can be obtained [7]. Other alloys of interest are those of Pd–Cu that show higher resistance to sulfur poisoning than pure palladium, which becomes irreversibly deactivated in the presence of very low concentrations of sulfur-containing streams [8–10].

To ensure high permeabilities, it is important to work with low membrane thickness without compromising membrane integrity. For this purpose, several techniques for the production of composite membranes, in which thin palladium alloy layers are deposited onto porous supports have been developed (Fig. 9.8) and are summarized by Drioli et al. [11]. The main problems related to composite membranes concern the achievement of defect-free deposited layers which maintain performance both with time, and also with thermal cycling. Usually, the dif-

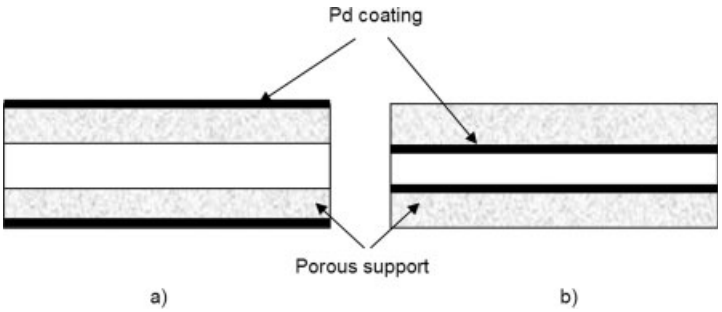


Figure 9.8 Composite membrane with a thin palladium alloy layer deposited onto the external (a) and onto the inner (b) surface of a tubular porous support.

ference in coefficients of thermal expansion between the support and the coating makes it difficult to avoid damage of the layer structure during use. Interfacial barrier layers can be used to limit the instability of the membrane layer/support interface.

Recently, a rolling technique has been proposed by the ENEA-Euratom Research Center of Frascati, Italy in which a metal ingot is processed by repeated cold rolling and annealing steps to reach the desired thickness [12]. In tubular reactors, the metal sheet is wound around the support to form two concentric tubes, avoiding direct contact between them. The lowest Pd thickness achieved by these methods is 50 microns.

9.3
Factors Influencing the Economics

When a reaction is carried out in traditional reactors (for example, fixed bed reactors), the main factors influencing the economics are the feed and process pressures (compressors), feed composition, flow rates, operating temperatures, product pressure, product purity and yield, flexibility, and capability for future expansion (see Table 9.1).

Factors involved in the economics of membrane reactors include those related to traditional reactors as well as parameters typical of the membrane device. First, the membrane type plays an important role. In particular, its permeability and selectivity affect, respectively, the capacity of the process and the product purity and, therefore, the membrane area needed for reaching a specific target.

The membrane can usually withstand a maximum temperature and pressure which can vary depending on membrane material and properties, thus influencing the choice of operating conditions. The choice of operating temperatures and pressures must also be made with consideration as to how they affect the permeability of species through the membrane. As previously discussed, the establishment of a partial pressure difference between the feed and the permeate side can

Tab. 9.1 Main factors influencing the economics of traditional reactors.

Feed pressure
Feed composition
Flow rates
Operating temperatures and pressures
Product pressure
Product purity
Product yield
Flexibility
Capability for future expansion

be obtained by pressurizing the feed side, using a sweep gas, or by applying vacuum to the permeate side. These factors all affect the economics. The recovery of the product at the permeate side at low pressure usually implies that the product must be compressed before further use (Fig. 9.9). The use of sweep gas dilutes the permeating product and implies an additional separation step downstream (Fig. 9.10).

Sometimes by-products generated during the reaction can foul the membrane surface with consequent reduction in membrane performance. Membrane performance can also be adversely affected by contaminants in the feed stream. In this case, a pre-treatment step before the membrane reactor becomes necessary for efficient operation.

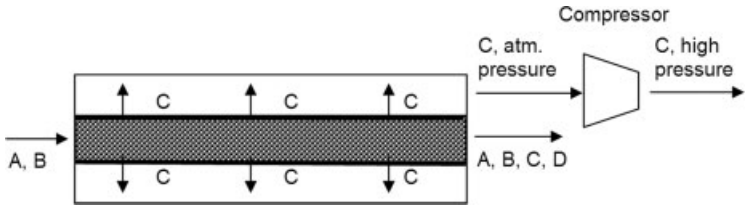


Figure 9.9 Compression of the product at the permeate side.

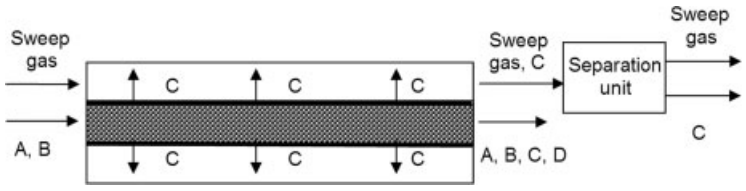


Figure 9.10 Separation of the product from the sweep stream.

Tab. 9.2 Main factors influencing the economics of membrane reactors.

Feed pressure
Feed composition
Flow rates
Operating temperatures and pressures (chosen taking into account also the membrane properties)
Product pressure
Product purity
Product yield
Membrane permeability
Membrane selectivity
Method used for establishing the driving force across the membrane
Membrane fouling
Lifetime of the membrane
Flexibility
Capability for future expansion

Membrane lifetime is another crucial aspect which must be taken into account because it determines the frequency of membrane replacement. Table 9.2 summarizes the main factors affecting the economics of a membrane reactor.

9.4 Dense Membrane Reactors for the Water-Gas Shift Reaction

The water-gas shift reaction is an exothermic equilibrium-limited reaction in which steam and carbon monoxide react to yield carbon dioxide and hydrogen:

$$CO + H_2O(g) = CO_2 + H_2 \tag{9.1}$$

The recovery of hydrogen at the permeate side of a membrane reactor permits:

1. Avoiding the presence of a separation unit downstream.
2. Shifting the reaction towards the right side (towards products).

Reaction shift towards products can be represented as a sequence of the following four steps:

1. Product formation and establishment of equilibrium,
2. Hydrogen removal,

3. Further product formation,
4. New equilibrium reached.

$$\underset{\text{1. } \textit{and} \text{ 4.}}{CO + H_2O \rightarrow CO_2 + H_2} \qquad \underset{\text{2.}}{CO + H_2O \rightarrow CO_2 + H_2\uparrow} \qquad \underset{\text{3.}}{CO + H_2O \rightarrow CO_2 + H_2} \tag{9.2}$$

The continuous removal of hydrogen through a selective membrane might lead to complete conversion of carbon monoxide.

Depending on the type of catalyst adopted, the operating temperatures can vary between 600 K (low-temperature shift catalyst) and 700 K (high-temperature shift catalyst). Because of the relatively high temperatures employed, studies reported in the literature usually refer to the use of inorganic membrane reactors [13–16]. In particular, because palladium and its alloys are extremely selective to hydrogen, many reports discuss work on palladium-based membrane reactors.

As mentioned, generally the use of Pd-based membranes implies, as a constraint, that the stream fed to the reactor does not contain sulfur. Palladium can also interact with other species involved in the water-gas shift reaction. For example, a reduction of hydrogen permeation can occur because of the adsorption of CO and steam onto the palladium surface, but generally the phenomena are limited to operating temperatures below 250 °C [17].

Both theoretical and experimental studies have been performed on palladium-based membrane reactors for the water-gas shift reaction. Ma and Lund simulated the performance achievable in a high temperature water-gas shift membrane reactor using both ideal membranes and catalysts [18]. By comparing the results obtained with those related to the existing palladium membrane reactors, they concluded that better membrane materials are not needed, and that higher performances mainly depend on the development of a water-gas shift catalyst not inhibited by CO_2. Marigliano et al. pointed out how the equilibrium shift conversion in membrane reactors is an increasing function of the sweep factor (defined as the ratio between the flow rate of the sweep at the permeate side and the flow rate of CO at the reaction side) [19]. The ratio is an index of the extractive capacity of the system.

Due to hydrogen transport through the membrane, the permeate becomes richer in hydrogen and, if a rapid removal is not achieved, counter-diffusion from the permeate to the reaction side can occur. Moreover, the hydrogen partial pressures established at the reaction and the permeate sides depend on whether the system operates under co-current or counter-current conditions and may be different. Criscuoli [20] analyzed the hydrogen profiles and the CO conversions achievable when working in, respectively, co-current and counter-current flow modes under the same operating conditions for two different feed compositions (Mix 1: CO 32%, CO_2 12%, H_2 4%, N_2 52%; Mix 2: CO 12.27%, CO_2 11.49%, H_2 75%, CH_4 1.24%). The membrane reactor considered was Pd /Ag-based (effective length, 15 cm; membrane diameter, 8 mm) with catalyst packed in the lumen. Figures 9.11 and 9.12 show the results. When the mixture fed to the membrane reactor is highly rich in hydrogen (Mix 2), hydrogen permeation occurs always

from the lumen (reaction side) to the shell (permeate side) for both configurations, and the conversions are quite similar. On the contrary, when working with a less hydrogen-rich feed (Mix 1), the hydrogen partial pressure at the permeate side is higher than that at the reaction side in the first 7 cm of the reactor for the counter-current flow mode, and counter-diffusion occurs, with a consequent decrease in the CO conversion. It is important to point out that the results achieved are related to the particular adopted operating conditions and can be different if, for example, the sweep flow rate or the feed pressure is increased, a membrane with different permeability is considered, or a different composition is fed to the reactor.

Conversions achievable in membrane reactors are quite high. For example, Uemiya et al. obtained a CO conversion of about 90% using a 20 micron layer of Pd coating on a porous glass tube [21]. The reaction was carried out on a commercial Fe–Cr high-temperature catalyst at 673 K with a ratio H_2O/CO of 2.

Basile et al. performed experimental studies on a composite palladium membrane obtained by coating an ultrathin double layer of palladium onto the inner surface of a commercial tubular ceramic membrane [22]. The membrane showed

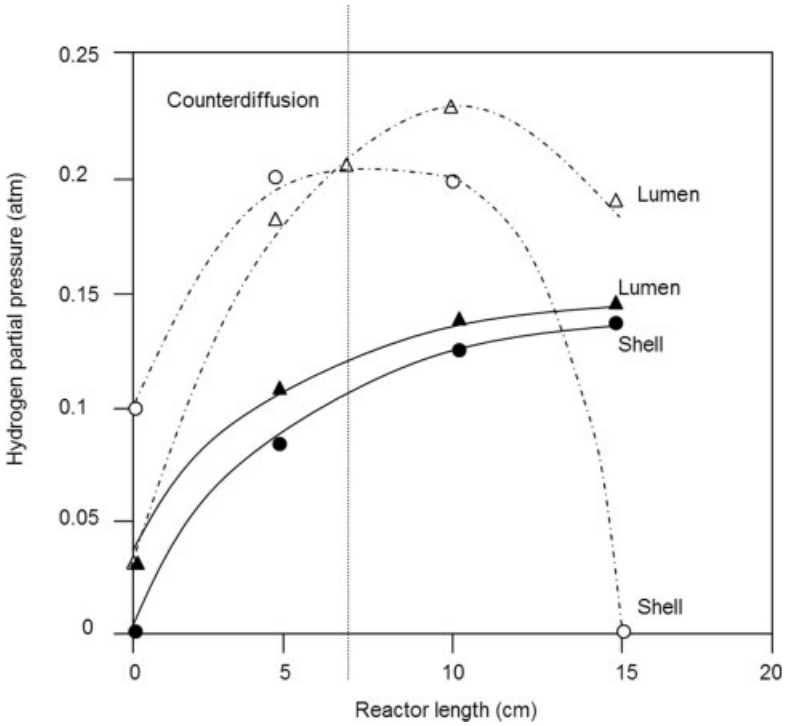


Figure 9.11 Hydrogen partial pressure profiles at the shell and lumen sides for co-current (solid symbols) and counter-current (open symbols) flow modes. Co-current conversion, 84.72%. Counter-current conversion, 80.63%. Operating conditions: Mix 1: *T*, 595 K; sweep, 2.97×10^{-5} mol s^{-1}; H_2O/CO, 1:1.

a separation factor, H_2/N_2, of 8.24 at 595 K, indicating that the coating was not defect free. Tests were carried out with a low-temperature shift catalyst which was packed in the lumen of the membrane. Two feed compositions were used: a mixture of reactants only, and a mixture containing reactants, products and nitrogen (53%). For the former mixture, by properly adjusting operating conditions, a maximum CO conversion of 99.89% was obtained, while, in all the investigated ranges, the CO conversions for the latter mixture never overcame equilibrium values. The authors attributed this result to competitive nitrogen permeation through the membrane.

The effect of feed composition on CO conversion has been studied by Criscuoli et al. in a defect-free tubular Pd-based membrane reactor with a low-temperature shift catalyst packed inside the lumen [23]. The three mixtures tested are summarized in Table 9.3. These authors found that for all feeds it was possible to overcome equilibrium conditions, and that the highest conversion was obtained when the composition of the mixtures was closer to that thermodynamically more favorable (complete conversion was achieved for Mix 1). In the same work, the authors performed system modelling and found that, with respect to traditional reactors,

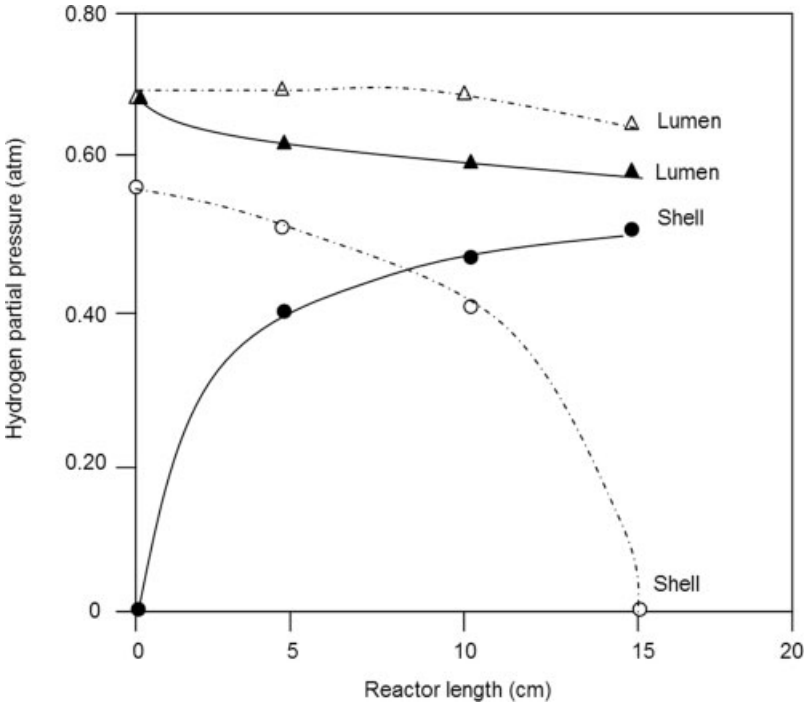


Figure 9.12 Hydrogen partial pressure profiles at the shell and lumen sides for co-current (solid symbols) and counter-current (open symbols) flow modes. Co-current conversion, 54.84%. Counter-current conversion, 55.60%. Operating conditions: Mix 2: *T*, 595 K; sweep, 2.97×10^{-5} mol s^{-1}; H_2O/CO, 1:1.

Tab. 9.3 Composition (on a dry basis) of the three mixtures used [23].

Mixture	CO (%)	CO_2 (%)	H_2 (%)	N_2 (%)	CH_4(%)
Mix 1	32.00	12.00	4.00	52.00	–
Mix 2	12.27	11.49	75.00	–	1.24
Mix 3	27.00	–	73.00	–	–

different kinetic expressions must be employed to describe membrane reactor behavior.

Tosti et al. proposed the use of Pd/Ag-based membrane reactors for the recovery of hydrogen and its isotopes from tritiated water in a "closed loop process" that includes both the forward and reverse water-gas shift reactions [24]. The aim of this process was to avoid any production of tritiated wastes and any consumption of CO. In the system, the retentate stream, rich in CO_2, was recycled to the reactor. The water-gas shift stops when all the water reacts and all the hydrogen is recovered at the permeate side. Then, hydrogen is added, and the CO_2 in the stream is completely converted to CO by the reverse water-gas shift reaction, thanks to the continuous removal of the produced water in a condenser. Figure 9.13 shows the proposed process.

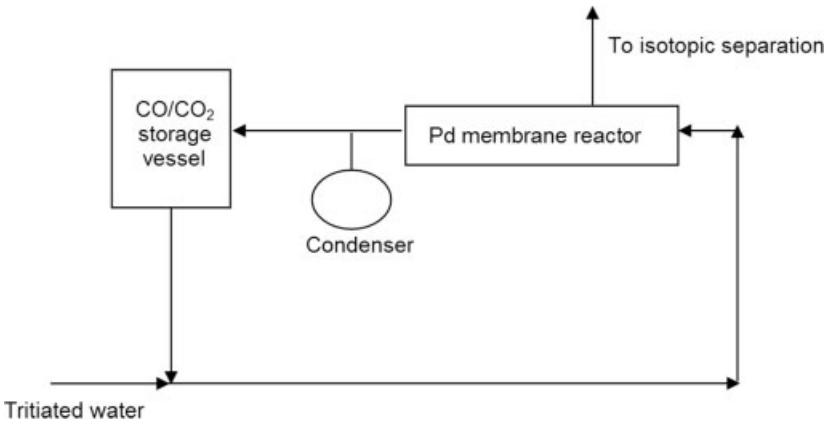


Figure 9.13 A schematic flow-sheet of the closed loop process.

9.5 Economic Feasibility of Water-Gas Shift Pd-based Membrane Reactors

The use of a membrane inside a reaction device has costs associated with the overall process. Specifically, palladium is an expensive material and, therefore, its cost can significantly affect the process economics.

In addition to the development of techniques able to produce membranes with low palladium thickness, a proper module design and module integration are the basis of an economics optimization. The module design is a factor of great importance for both membrane and traditional reactors because it determines fluidodynamics inside the device (mass transfer efficiency) as well as pressure drops. However, in membrane reactors, the performance and, therefore, associated economics, is more dependent on the module design. In membrane systems, it is important to optimize conditions not only on the feed side but also on the permeate side. The fluidodynamics established in the reaction zone must allow the desired permeation rate through the membrane as well as the proper residence time of the reactants inside the module. Low partial pressures at the permeate side must be ensured (this implies that the permeated species must be rapidly removed from the permeate side). Moreover, in order to guarantee a constant difference of partial pressure across the membrane for all its length (a constant permeation rate), careful control of the pressure drops in the membrane reactor becomes fundamental. The pressure drops in the module also influence energy consumption.

The system design must address the performance target to be reached (such as a particular conversion value or hydrogen recovery, etc.). For example, if both conversion and separation must be the highest possible, then a large membrane area is needed. On the contrary, if lower yields are acceptable, lower membrane areas are required, but the retentate stream needs to be further treated. A comparison between the costs associated with the two cases leads, usually, to the final design.

Optimization of operating conditions is also made with reference to the final target. The choice of whether to pressurize the feed stream or to use vacuum or a sweep gas on the permeate side, depends on associated costs and on the specific case considered. For example, if the feed stream is already available in the plant at the pressure needed, the permeate can operate at atmospheric pressure. Otherwise, the energy required to apply vacuum or to pressurize the feed must be compared to the costs of the sweep gas as well as the costs related to the additional separation of the diluted permeate.

Module assembly, whether in series or in parallel, is also dictated by the particular goal to be reached. The arrangement in parallel leads to higher capacity, while the series arrangement ensures higher yields (but higher pressure drops). In choosing the final flow sheet, the higher system performance must be compared to membrane area costs, compression, and other relevant parameters.

By coupling the membrane reactor to other units, it is possible to improve its performance. For example, Bracht et al. [14] considered two-step adiabatic reactors instead of a one-step isothermal reactor as a means of extending membrane life-

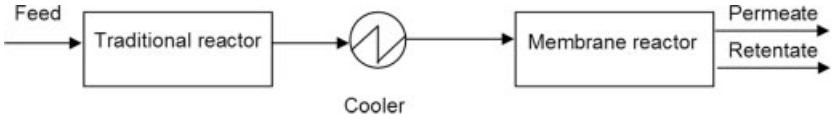


Figure 9.14 Coupling of traditional and membrane reactors.

time (Figure 9.14). In the first (conventional) reactor, the CO conversion accompanied by the largest temperature rise occurs, then the cooled exit stream further reacts in the membrane reactor. In this way the temperature rise in the membrane reactor is moderate, and the lifetime of the membrane can be improved.

When a plant is in operation, different factors can vary (for example, feed composition, energy cost, etc.), and the ability to handle changes without modifying the plant design plays a significant role in the process economics. For this reason, in defining the optimal membrane system design, it is also important to perform some "sensitivity analysis" aimed at studying the effect that various parameters can have on the overall performance and cost. Table 9.4 lists some of the factors that can change with time.

Costs associated with a membrane reactor can be divided into capital and operating costs.

Capital costs include mainly equipment and catalyst costs, installation costs, and indirect costs such as construction overhead and contingency. Operating costs are related mainly to energy consumption, raw materials, membrane replacement, etc. Tables 9.5 and 9.6 summarize the main factors that influence capital and operating costs.

Tab. 9.4 Some of the factors that can vary with time.

Feed pressure
Feed composition
Feed flow rate
Product purity requirements
Product yield requirements
Plant capacity
Membrane (Pd) cost
Membrane lifetime
Membrane performance
Raw materials cost
Energy cost
Labor cost

Tab. 9.5 Main factors included in capital costs.

Land
Equipment
Catalyst
Installation
Transportation
Site development (roads, buildings, etc.)
Construction overhead and contingency

Tab. 9.6 Main factors included in operating costs.

Raw materials
Energy
Membrane replacement
Replacement parts
Labor
Maintenance

Tab. 9.7 Variation of the Pd cost with time [25].

Month (2002)	**Pd Cost (Euro/g)**
January	15.7
March	14.0
May	13.0
July	10.3
September	10.8

The cost of palladium is an important factor influencing the economics of Pd-based membrane reactors. Table 9.7 shows how the cost of palladium varied in 2002 [25].

Little discussion is available in the literature on membrane reactor economics [14, 26–28].

As a particular example, an economic study carried out by Criscuoli et al. [29] for the water-gas shift reaction in a Pd-based membrane reactor is discussed below.

In industrial plants, the water-gas shift reaction is carried out in two reactors in series (high-temperature (HT) and low-temperature (LT) water-gas shift reactors). The hydrogen produced at the reaction device exit is recovered in a separation unit downstream (Fig. 9.15). In the study performed, the two reactors plus the separator were compared in terms of capital and operating costs, with a Pd-based membrane reactor in which the pure hydrogen is recovered at the permeate side (Fig. 9.16). Table 9.8 shows the main operating conditions of the industrial plant considered. The membrane reactor unit was designed in order to operate with industrial quantities and to achieve the same industrial hydrogen recovery. An integrated membrane system in which the feed stream is first fed to a Pd-based

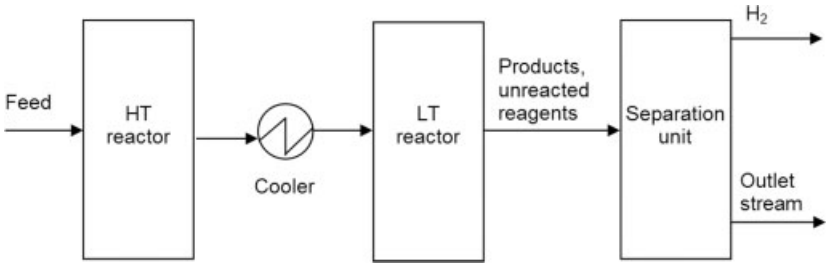


Figure 9.15 Conventional system used for carrying out the water-gas shift reaction.

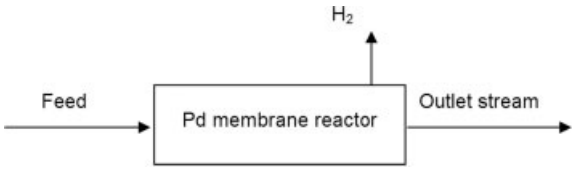


Figure 9.16 A Pd-based membrane reactor for carrying out the water-gas shift reaction.

Tab. 9.8 Main operating conditions of the industrial plant [29].

Temperature (K)	623
Pressure (atm)	16
H_2O/CO	9.8
Inlet flow rate (kmol h^{-1})	461.57
Pure hydrogen recovered (kmol h^{-1})	138.00

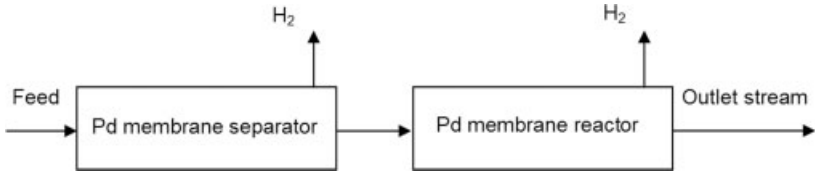


Figure 9.17 A Pd-based membrane separator integrated with a Pd-based membrane reactor.

membrane separator, in which hydrogen present in the feed mixture is partially removed and then sent to the Pd-based membrane reactor, has also been designed (Fig. 9.17) and compared to the other systems.

For both membrane systems investigated (Pd membrane reactor, and Pd membrane separator plus Pd membrane reactor), the design was carried out by operating at the same reactant molar ratio used in the industrial plant (Pd Reactor 1) as well as lower values (Pd Reactor 2). The feed stream is available in the plant at 16 atm (16.2 bar) and, therefore, no further compression of the feed was considered for the membrane units. Furthermore, no sweep gas or vacuum was applied at the permeate side, and the hydrogen coming from the membrane units was pure, without any need for separation devices downstream. However, the hydrogen pressure was too low to be directly used and, therefore, a compression step has been taken into account.

The main operating conditions and characteristics of membrane devices are summarized in Table 9.9.

Table 9.10 shows the comparison between the membrane and the industrial systems in terms of module characteristics and costs. All membrane systems investi-

Tab. 9.9 Main operating conditions and characteristics of the membrane devices [29].

	Pd Reactor 1	Pd Reactor 2	Pd Separator/ Pd Reactor 1	Pd Separator/ Pd Reactor 2
Temperature (K)	595	595	595/595	595/595
Pressure at the lumen side (atm)	16	16	16/16	16/16
Pressure at the shell side (atm)	1	1	1/1	1/1
H_2O/CO	9.8	2	No water/9.8	No water/2
Inlet flow rate (kmol h^{-1})	461.57	246.00	190.80/ 354.80	190.80/ 139.27
Pure hydrogen recovered (kmol h^{-1})	138.00	138.00	107.00/31.00	107.00/31.00
Pd thickness (micron)	75	75	75	75

Tab. 9.10 Comparison of membrane and industrial systems [29].

	Industrial Plant	Pd Reactor 1	Pd Reactor 2	Pd Separator/Pd Reactor 1	Pd Separator/Pd Reactor 2
Volume (m^3)	10.72	5.97	2.28	1.43/2.47	1.43/0.63
Catalyst (kg)	5500	4885	1866	0/2021	0/515
Membrane area (m^2)	–	1193	456	285/494	285/126
Capital costs (Euro)	1.16×10^6	6.78×10^6	2.82×10^6	4.57×10^6	2.59×10^6
Operating costs (Meuro y^{-1})	2.43	6.97	3.72	5.34	3.55

gated allowed lower catalyst volume and amount, but they are more expensive than the industrial ones, due essentially to the high cost of palladium. A way to reduce this cost is to work with a lower palladium thickness (less palladium is used to produce the membrane, and less membrane area is required for the same hydrogen recovery). In particular, the authors found that defect-free palladium membranes with a thickness of 20 microns or less are needed in order to make Pd-based membrane reactors competitive with traditional systems.

9.6 Future Directions

The potential of membrane reactors has been widely verified and documented for a large number of reactions. However, all the studies made are still confined to the "laboratory scale," and their implementation in industrial systems has yet to occur. Research into new membrane materials and improvement in the properties of currently available membranes (permselectivity, resistance to poisoning, stability, reduction of palladium thickness, etc.) are always in progress. The development of procedures to deposit the catalyst within the membrane structure without changing its initial permeability and selectivity is an example of the ongoing research for the preparation of catalytic membranes.

Although these types of studies are important for further improving the efficiency of processes and extending the field of applications, the implementation of membrane reactors on a large scale also needs to be addressed.

In industrial applications, membrane reactors will undergo repeated thermal cycles and will need to work at high temperatures and pressures in aggressive chemical environments over a number of years. Good seals between membrane and module hardware must be achieved in order to avoid problems of mixing streams of permeate and retentate or by-passing during operation. The membrane, as well as the catalyst, must withstand industrial operating conditions.

Therefore, data on long term research under realistic industrial conditions are needed.

Kinetic studies devoted to better understanding the influence of membranes on reaction kinetics and the interactions between membrane and catalyst represent other issues. For example, Dixon reported some changes in catalyst behavior due to exposure to a different reaction mixture composition resulting from adding or removing species through the membrane [30]. Moreover, from some first modelling results, it appears that kinetic expressions to be used in membrane reactors are different from those of conventional reaction systems [23]. The development of mathematical models to be used for design optimization and for easy scale-up is also very important for a rapid evaluation of the effective size of the membrane system and of all the chemical feeds (and their properties) involved in a specific application.

In order to be used on an industrial scale, membrane reactors need to be produced at a scale much larger than those currently employed in laboratory set-ups. This implies that membrane fabrication methods for large scale systems need to be defined, as well as the engineering related to module preparation.

Engineering studies will help in producing membrane reactors with high membrane area per unit volume and in optimizing their performance. Furthermore, systems of control for large scale modules are required.

The cost associated with membrane reactors is, evidently, the "key factor" for their implementation on an industrial scale. Inorganic membranes can be expensive (especially Pd-based membranes) and, generally, their cost strongly affects the total costs of the process. However, cost data available often refer to membranes used in the laboratory, and it is difficult to obtain reliable information on the cost of production for large scale, because, as already mentioned, the technology of production of large scale membrane reactors is not yet well defined.

Based upon the above considerations, it is concluded that the implementation of membrane reactors at an industrial level will be strongly related to further development, testing and optimization of larger scale membrane reactors under real operating conditions, in order to obtain effective performance and cost data to be directly used for techno-economic evaluations by industrial companies.

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